Exercise 4

Task 1

a) In the s-plane $j\omega = s = 2f_s \frac{z-1}{z+1}$

$$\Rightarrow H_{LPF}(z) = \frac{1}{1 + \frac{2f_s \frac{z-1}{z+1}}{2\pi f_0}} = \frac{2\pi f_0 (z+1)}{2\pi f_0 (z+1) + 2f_s (z-1)}$$

$$= \frac{2\pi f_0 \frac{(z+1)}{z}}{(2\pi f_0 + 2f_s) + (2\pi f_0 - 2f_s) z^{-1}}$$

$$= \frac{\frac{2\pi f_0}{(2\pi f_0 + 2f_s)} (1 + z^{-1})}{1 + \frac{(2\pi f_0 - 2f_s) z^{-1}}{(2\pi f_0 + 2f_s)}}$$

$$\Rightarrow B = \frac{2\pi f_0}{2\pi f_0 + 2f_s} = \frac{f_0}{f_0 + f_s/\pi}$$

$$A_1 = \frac{2\pi f_0 - 2f_s}{2\pi f_0 + 2f_s} = \frac{\pi f_0 - f_s}{\pi f_0 + f_s}$$

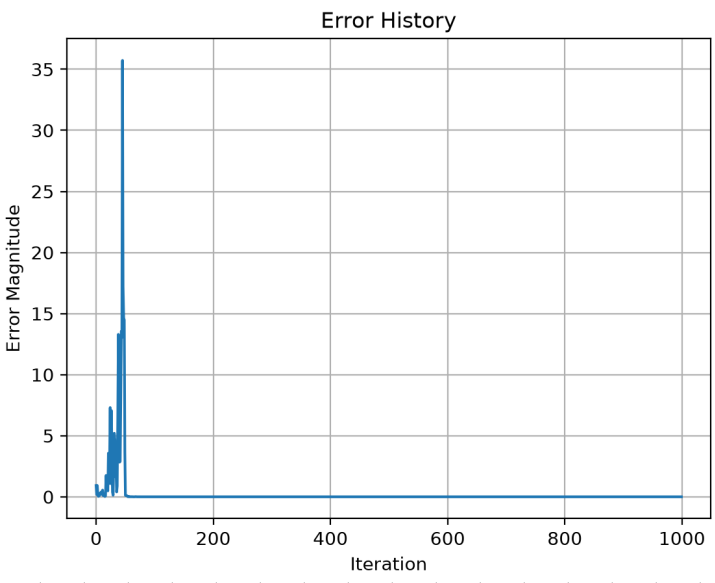
$$H_{LPF1}(z) = B \cdot \frac{1 + z^{-1}}{1 + A_1 z^{-1}}$$

b) Check added figure.

Task 2

Code and generated plots are added in zip along with this pdf.

$\alpha \approx 0{,}5 \Rightarrow$ error $\to 0$



$\alpha \approx 5 \Rightarrow$ 1e307 scale = error exploded

